LOCALLY BOUNDED FLAT COMPLEXES OF MODULES OVER THE STRUCTURE SHEAF HAVE BOUNDED, QUASI-COHERENT, FINITE TOR DIMENSION PUSHFORWARDS THAT COMMUTE WITH BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Editorial note. One sentence inside the counted proof reads, verbatim: 'Some details omitted.' (at the end of the argument for part (5), where the author says the induction on the length of the bounded complex finishes after the distinguished triangle is displayed). The counted span is otherwise the author's own proof of all five parts, and the omitted clause concerns the induction whose displayed triangle and base case are given in the same span; per the project rule the phrase is recorded here verbatim.

Allowed standard prerequisites (editorial).

- The first spectral sequence of a complex of functors converges and relates the higher direct images of the terms to those of the complex. invoked at proof lines 5797-5801. Produces the local boundedness and quasi-coherence of the pushforward of the complex.
- Higher direct images of quasi-coherent modules under a quasi-compact quasi-separated morphism are quasi-coherent, and the biregular spectral sequence converges. invoked at proof lines 5799-5802. Supplies the quasi-coherence statement needed for parts (1) and (2).
- The induction principle on quasi-compact opens reduces a statement about all quasi-compact opens to the affine case, and relative Mayer-Vietoris plus the behaviour of tor amplitude in triangles handles the union step. invoked at proof lines 5803-5815. Reduces part (3) to the affine case, where the pushforward is computed by an acyclic resolution.
- For an affine morphism the higher direct images of quasi-coherent modules vanish, so the pushforward of the complex is the complex of the pushforwards. invoked at proof lines 5817-5820. Turns the pushforward in the affine case into a bounded complex of flat modules, giving finite tor dimension.
- The projection formula supplies a functorial comparison map between the two sides of the base change statements. invoked at proof lines 5827-5830.

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Reduces parts (4) and (5) to showing that the comparison map is an isomorphism.
- The pushforward of a quasi-coherent module commutes with direct sums, and the derived pullback of a quasi-coherent complex is computed by the tensor product. invoked at proof lines 5835-5845. Lets the proof check the commutation with direct sums that the cone argument needs for the single module case.

Source reference. Original lemma 015J: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 5797-5799 for the spectral sequence used to prove parts (1) and (2); the conclusion drawn from it is inside the counted span.

Source reference. Original lemma 01XJ: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 5799-5801 for the quasi-coherence input of part (2); the spectral sequence argument that consumes it is inside the counted span.

Source reference. Original lemma 08DR: sources/stacks-chow/tags.txt, source lines 1–1. Invoked at proof lines 5803-5805 to reduce part (3) to the affine case; the union step and the affine computation are inside the counted span.

Source reference. Original lemma 08HZ: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 5807-5810 for the displayed triangle used in the union step; the tor amplitude conclusion is inside the counted span.

Source reference. Original lemma 08CJ: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 5811-5813 to pass from the pieces of the triangle to the union; the application is inside the counted span.

Source reference. Original lemma 015E: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 5817-5819 for the identification of the pushforward of the complex with the complex of pushforwards in the affine case; the flatness discussion that follows is inside the counted span.

Source reference. Original lemma 01XC: sources/stacks-chow/tags.txt, source lines 1–1. Cited with the previous label at proof lines 5818-5820 for the same identification; the finiteness of tor dimension concluded from it is inside the counted span.

Source reference. Original equation 0B53: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 5827-5830 to obtain the comparison map whose bijectivity is the content of part (5); the reduction to the affine case and the direct sum argument are inside the counted span.

Source reference. Original lemma 08DE: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 5838-5841 to identify the tensor product in the single degree case; the commutation with direct sums drawn from it is inside the counted span.

Source reference. Original lemma 08DZ: sources/stacks-perfect/pinned-original-source.tex, source lines 803–803. Used at proof lines 5843-5845 for the single degree case of the direct sum argument; the induction that extends it to the whole complex is inside the counted span.

Source reference. Original lemma 08EU: sources/stacks-perfect/pinned-original-source.tex, source lines 5269–5269. The proof states at proof lines 5831-5834 that

the rest is the same as the proof of that statement except for the direct sum argument; the direct sum argument specific to this lemma is inside the counted span.

Source reference. Original lemma 012W: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 5799-5802 together with the quasi-coherence statement to conclude parts (1) and (2); the spectral sequence argument that consumes the convergence is inside the counted span.

Lemma 0.2. *Let $a : X \to S$ be a quasi-compact and quasi-separated morphism of schemes. Let $\mathcal{F}^\bullet$ be a locally bounded complex of $a^{-1}\mathcal{O}_S$-modules. Assume for all $n \in \mathbf{Z}$ the sheaf $\mathcal{F}^n$ is a flat $a^{-1}\mathcal{O}_S$-module and $\mathcal{F}^n$ has the structure of a quasi-coherent $\mathcal{O}_X$-module compatible with the given $a^{-1}\mathcal{O}_S$-module structure (but the differentials in the complex $\mathcal{F}^\bullet$ need not be $\mathcal{O}_X$-linear). Then the following hold*

(1) *$Ra_*\mathcal{F}^\bullet$ is locally bounded,*
(2) *$Ra_*\mathcal{F}^\bullet$ is in $D_{QCoh}(\mathcal{O}_S)$,*
(3) *$Ra_*\mathcal{F}^\bullet$ locally has finite tor dimension,*
(4) *$\mathcal{G} \otimes^{\mathbf{L}}_{\mathcal{O}_S} Ra_*\mathcal{F}^\bullet = Ra_*(a^{-1}\mathcal{G} \otimes_{a^{-1}\mathcal{O}_S} \mathcal{F}^\bullet)$ for $\mathcal{G} \in QCoh(\mathcal{O}_S)$, and*
(5) *$K \otimes^{\mathbf{L}}_{\mathcal{O}_S} Ra_*\mathcal{F}^\bullet = Ra_*(a^{-1}K \otimes^{\mathbf{L}}_{a^{-1}\mathcal{O}_S} \mathcal{F}^\bullet)$ for $K \in D_{QCoh}(\mathcal{O}_S)$.*

Proof. Parts (1), (2), (3) are local on S hence we may and do assume S is affine. Since a is quasi-compact we conclude that X is quasi-compact. Since $\mathcal{F}^\bullet$ is locally bounded, we conclude that $\mathcal{F}^\bullet$ is bounded.

For (1) and (2) we can use the first spectral sequence $R^p a_*\mathcal{F}^q \Rightarrow R^{p+q}a_*\mathcal{F}^\bullet$ of Derived Categories, Lemma 015J. Combining Cohomology of Schemes, Lemma 01XJ and Homology, Lemma 012W we conclude.

Let us prove (3) by the induction principle of Cohomology of Schemes, Lemma 08DR. Namely, for a quasi-compact open of U of X consider the condition that $R(a|_U)_*(\mathcal{F}^\bullet|_U)$ has finite tor dimension. If U, V are quasi-compact open in X, then we have a relative Mayer-Vietoris distinguished triangle

$$R(a|_{U\cup V})_*\mathcal{F}^\bullet|_{U\cup V} \to R(a|_U)_*\mathcal{F}^\bullet|_U \oplus R(a|_V)_*\mathcal{F}^\bullet|_V \to R(a|_{U\cap V})_*\mathcal{F}^\bullet|_{U\cap V} \to$$

by Cohomology, Lemma 08HZ. By the behaviour of tor amplitude in distinguished triangles (see Cohomology, Lemma 08CJ) we see that if we know the result for U, V, $U \cap V$, then the result holds for $U \cup V$. This reduces us to the case where X is affine. In this case we have

$$Ra_*\mathcal{F}^\bullet = a_*\mathcal{F}^\bullet$$

by Leray's acyclicity lemma (Derived Categories, Lemma 015E) and the vanishing of higher direct images of quasi-coherent modules under an affine morphism (Cohomology of Schemes, Lemma 01XC). Since $\mathcal{F}^n$ is S-flat by assumption and X affine, the modules $a_*\mathcal{F}^n$ are flat for all n. Hence $a_*\mathcal{F}^\bullet$ is a bounded complex of flat $\mathcal{O}_S$-modules and hence has finite tor dimension.

Proof of part (5). Denote $a' : (X, a^{-1}\mathcal{O}_S) \to (S, \mathcal{O}_S)$ the obvious flat morphism of ringed spaces. Part (5) says that

$$K \otimes^{\mathbf{L}}_{\mathcal{O}_S} Ra'_*\mathcal{F}^\bullet = Ra'_*(L(a')^*K \otimes^{\mathbf{L}}_{a^{-1}\mathcal{O}_S} \mathcal{F}^\bullet)$$

Thus Cohomology, Equation (0B53) gives a functorial map from the left to the right and we want to show this map is an isomorphism. This question is local on S hence we may and do assume S is affine. The rest of the proof is *exactly* the same as the proof of Lemma 08EU except that we have to show that the functor

$K \mapsto Ra'_*(L(a')^*K \otimes^{\mathbf{L}}_{a^{-1}\mathcal{O}_S} \mathcal{F}^\bullet)$ commutes with direct sums. This is where we will use $\mathcal{F}^n$ has the structure of a quasi-coherent $\mathcal{O}_X$-module. Namely, observe that $K \mapsto L(a')^*K \otimes^{\mathbf{L}}_{a^{-1}\mathcal{O}_S} \mathcal{F}^\bullet$ commutes with arbitrary direct sums. Next, if $\mathcal{F}^\bullet$ consists of a single quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}^\bullet = \mathcal{F}^n[-n]$ then we have $L(a')^*G \otimes^{\mathbf{L}}_{a^{-1}\mathcal{O}_S} \mathcal{F}^\bullet = La^*K \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{F}^n[-n]$, see Cohomology, Lemma 08DE. Hence in this case the commutation with direct sums follows from Lemma 08DZ. Now, in general, since S is affine (hence X quasi-compact) and $\mathcal{F}^\bullet$ is locally bounded, we see that

$$\mathcal{F}^\bullet = (\mathcal{F}^a \to \ldots \to \mathcal{F}^b)$$

is bounded. Arguing by induction on $b-a$ and considering the distinguished triangle

$$\mathcal{F}^b[-b] \to (\mathcal{F}^a \to \ldots \to \mathcal{F}^b) \to (\mathcal{F}^a \to \ldots \to \mathcal{F}^{b-1}) \to \mathcal{F}^b[-b+1]$$

the proof of this part is finished. Some details omitted.

Proof of part (4). Let $a' : (X, a^{-1}\mathcal{O}_S) \to (S, \mathcal{O}_S)$ be as above. Since $\mathcal{F}^\bullet$ is a locally bounded complex of flat $a^{-1}\mathcal{O}_S$-modules we see the complex $a^{-1}\mathcal{G} \otimes_{a^{-1}\mathcal{O}_S} \mathcal{F}^\bullet$ represents $L(a')^*\mathcal{G} \otimes^{\mathbf{L}}_{a^{-1}\mathcal{O}_S} \mathcal{F}^\bullet$ in $D(a^{-1}\mathcal{O}_S)$. Hence (4) follows from (5). $\square$